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Question_1:

Question description

Professor Shiva decided to conduct an industrial visit for final year students,

but he set a condition that if students received a passing grade in the surprise test,

they would be eligible to go on the industrial visit.

He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test.

Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked.

Input Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Next line indicates the node data D that has to be deleted.

Output Format

Single line represents the linked list after required elements deleted.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
7 9 8 8 4 2
8
```

EXPECTED OUTPUT

```
Linked List:->7->9->4->2
```

Test Case 2

INPUT (STDIN)

```
10
5 3 7 2 7 9 8 8 4 2
2
```

EXPECTED OUTPUT

```
Linked List:->5->3->7->7->9->8->8->4
```

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Code:

```cpp
/*
QUESTION_1:
Given a linked list and a data value D, delete all the nodes
containing D from the linked list and display the final linked list.
*/
#include <iostream>
using namespace std;

struct node {
    int data;
    node *next;
};
void create(node *&head, int x) {
    node *p = new node{x, NULL};

    if (!head)
        head = p;
    else {
        node *t = head;
        while (t->next)
            t = t->next;
        t->next = p;
    }
}
void del(node *&head, int x) {
    while (head && head->data == x)
        head = head->next;
    node *p = head;

    while (p && p->next) {
        if (p->next->data == x)
            p->next = p->next->next;
        else
            p = p->next;
    }
}
int main() {
    int n, x;
```

```cpp
    cin >> n;

    node *head = NULL;
    for (int i = 0; i < n; i++) {
        cin >> x;
        create(head, x);
    }
    cin >> x;
    del(head, x);

    cout << "Linked List:";
    while (head) {
        cout << "->" << head->data;
        head = head->next;
    }
    return 0;
}
```

Output:

Test Case_1:

```
6
7 9 8 8 4 2
8
Linked List:->7->9->4->2
PS C:\Users\Prabin Yadav\D
```

Test Case_2:

```
10
5 3 7 2 7 9 8 8 4 2
2
Linked List:->5->3->7->7->9->8->8->4
PS C:\Users\Prabin Yadav\Documents\III
```